Multi-Domain Emotion Recognition Enhancement: A Novel Domain Adaptation Technique for Speech-Emotion Recognition

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***Abstract*—As artificial intelligence advances, speech-emotion recognition (SER) has become a critical research area. Traditional SER methods typically rely on homogeneous domain data for training and testing. This practice requires adaptation when confronted with real-world data's heterogeneous linguistic, methodological, and speaker-related attributes. These variances can degrade the accuracy and generalization of SER models. To address this gap, we introduce a novel domain adaptation technique, multi domain emotion recognition enhancement (MDERE), which utilizes a non-negative matrix to reduce the inflexibility of the conventional binary label matrix for source domain data. This process yields a label matrix that better adapts to the nuances of the source labels while preserving their original structure. This framework refines SER methods by fine-tuning a transformation matrix for enhanced emotion discrimination. Elastic net regularization, which combines L1 and L2 penalties, enriches the transformation matrix, selectively emphasizing relevant features to enhance the robustness of emotion detection. The framework constructs customized similarity and dissimilarity graphs to reconcile the differences between source and target domains, enabling nuanced cross-domain data analysis. Extensive testing on multiple cross-domain SER tasks has shown that MDERE substantially improves recognition accuracy, surpassing several state-of-the-art algorithms. These results demonstrate MDERE's ability to effectively align domain variations enhances the generalizability of SER systems.**

***Index Terms*—Domain adaptation techniques, feature transformation matrix, label matrix refinement, non-negative matrix factorization, speech emotion recognition.**

Received 17 December 2023; revised 21 August 2024 and 27 September 2024; accepted 1 November 2024. Date of publication 14 November 2024; date of current version 27 January 2025. This work was supported in part by the Higher Education Sprout Project, Ministry of Education, in part by the Yushan Young Scholar Program, Ministry of Education and National Yang Ming Chiao Tung University (NYCU) under Grant 113C51N042, in part by the Healthy Longevity Global Grand Challenge Catalyst Award, U.S. National Academy of Medicine and Academia Sinica under Grant AS-HLGC-111-02, and in part by the National Science and Technology Council under Grant NSTC 112-2221-E-A49-160-MY3. The associate editor coordinating the review of this article and approving it for publication was Prof. Joseph Keshet.*(Corresponding author: Li-Chia Tai.)*

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Digital Object Identifier 10.1109/TASLP.2024.3498694

I. Introduction

IN THE current landscape of artificial intelligence research, speech-emotion recognition (SER) has become a pivotal field of study. SER systems can interpret speech signals and categorize the emotional states conveyed by speakers, identifying fundamental emotions such as anger, happiness, fear, boredom, disgust, sadness, and the neutral state [1], [2]. This provides a path toward emotion and mental health monitoring, distinguishing it from sensors designed for physiological health tracking [3]. The application of SER is extensive, finding significance in intelligent home robotics, interactive computer tutorials, vehicular systems, call centers, and mobile communications [4], [5]. Despite these valuable applications, SER systems often need more consistent performance across different domains due to data variability, a challenge that necessitates innovative solutions. Advances in pattern recognition have ushered in a host of sophisticated algorithms applied to SER, ranging from support vector machines (SVM) [6], multi-task learning [7], [8], convolutional neural networks (CNN) [9], [10], long short-term memory (LSTM) networks [11], to regression-based methods [12], [13]. However, a prevalent issue in practical scenarios is the divergence in training and testing data, attributed to variations in language, background noise, and data collection circumstances. This domain discrepancy significantly hampers the performance of SER methods. In light of these diverse applications, robust SER systems that perform well across varied domains are imperative, a challenge multi-domain emotion recognition enhancement (MDERE) is designed to meet. This domain discrepancy leads to a performance gap, which domain adaptation seeks to bridge.

Domain adaptation (DA) has been identified as an efficacious strategy to mitigate this issue by aligning the domains and minimizing distribution discrepancies. A multitude of DA techniques have been proposed in the last few decades. For instance, In [14] authors introduced a joint distribution adaptation (JDA) method that aligns the marginal and conditional distributions between the source and target domains to facilitate knowledge transfer. The use of low-rank and sparse representation for superior feature discrimination has been explored [15]. Effectiveness in

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the context of SER tasks has been validated, mainly through methods like domain-adaptive least squares regression, which calculates an optimal projection matrix using both source and target data, demonstrating impressive efficacy [16]. However, the methods above directly utilize the one-hot encoded matrix as the regression matrix, which can be overly restrictive and adversely impact regression performance [17]. A one-hot encoded matrix is a binary matrix where each row corresponds to a class label, and each column represents a sample. A '1' indicates the presence of a specific class label for a given sample, and a '0' indicates its absence. While this straightforward binary labeling method is widely used, it may only partially capture the nuanced relationships between samples and classes. This limitation can hinder the performance of regression-based methods, particularly in cross-domain scenarios [18], [19]. Several recent studies have focused on applying domain adaptation techniques specifically for SER, addressing the challenges of domain discrepancies. For instance, Abdelwahab and Busso [20] introduced a domain adversarial approach for acoustic emotion recognition, highlighting the effectiveness of adversarial training in reducing domain mismatches. Chao et al. [21] explored cross-corpus speech emotion recognition by leveraging few-shot learning and domain adaptation, demonstrating significant improvements in scenarios with limited target domain data. Given these advancements, our proposed MDERE framework offers a novel approach by integrating similarity and dissimilarity graph-based strategies, further enhancing cross-domain emotion recognition performance. This approach helps to reduce domain discrepancies, thereby improving the generalization of SER systems across varied datasets. To overcome these restrictions, MDERE employs a non-negative matrix approach and elastic net regularization (ENR), promoting a model that is both flexible and discriminatively powerful, unlike its predecessors. A progressively discriminative transfer network (PDTN) was introduced in response to this limitation [22]. PDTN incorporates clustering information for each category as prior knowledge and introduces a multi-layer distance metric known as the maximum mean discrepancy (MMD) for measuring distributional differences [22]. It is worth noting that while MMD is effective at capturing differences in mean values within the feature space, it may overlook the intrinsic structural information of the data, which is crucial for knowledge transfer [23]. Given the discussed literature, our contribution is the MDERE framework for SER across different domains. Our approach utilizes a label relaxation strategy to replace the zero-one label matrix, resulting in a more flexible model. We also design similarity and dissimilarity graphs to bridge the domain gap. Furthermore, we incorporate an ENR into the transformation matrix to highlight discriminative features [24]. Fig. 1 illustrates the MDERE framework. These contributions are substantiated in the latter sections, culminating in empirical validation of MDERE's efficacy. MDERE introduces a unique domain adaptation strategy that not only accounts for the limitations of current approaches but also introduces innovative mechanisms for enhanced adaptability and accuracy in SER.

The main contributions of our approach are as follows:

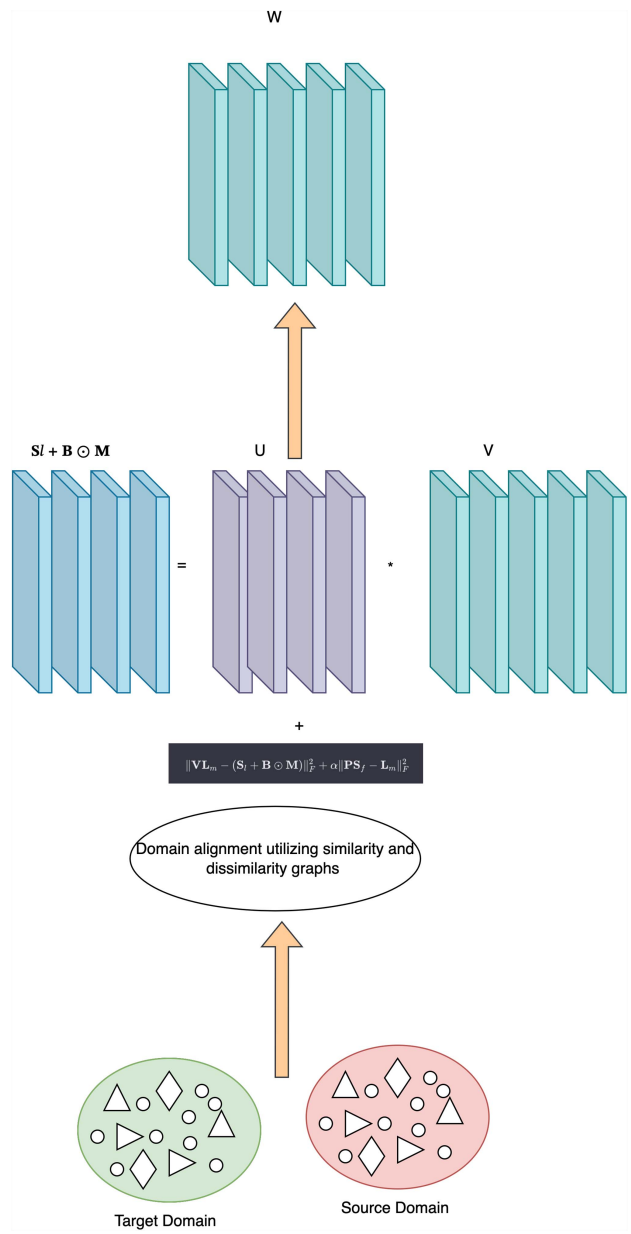

Fig. 1. An overview of the MDERE model architecture. The figure demonstrates the alignment process between source and target domains using similarity and dissimilarity graphs.

- We propose the MDERE framework, a domain-adaptive dual-relaxation regression technique designed for narrowing the domain gap in SER across various domains.
- A dual-relaxation regression strategy is developed, utilizing a relaxed label matrix for regression to effectively guide the transfer learning process.
- We introduce a new distance metric that meticulously explores the relationships of similarity and dissimilarity between samples from different domains.

The remainder of this paper is structured as follows: Section II presents an overview of the related work in linear regression and DA methods. Section III describes our proposed MDERE methodology and its optimization algorithm. Experimental

setups and the analysis of results are discussed in Section IV. Finally, we conclude our findings in Section V.

II. Related Works

This section briefly recalls the works related to our proposed method, including linear regression (LR) and DA.

A. Linear Regression

Linear methods have been widely used in various tasks such as image classification [25], [26], [27], and SER problems [16], [28], among others. These methods use a transformation matrix to map features to the label space. The traditional linear regression model is expressed as:

$$\min_{W} \|Y - WX\|_F^2 \tag{1}$$

In (1), $\mathbf{X}$ represents the input data matrix with dimensions $d \times n$, where d is the number of features and n is the number of samples. Each column corresponds to a feature vector, and each row corresponds to an observation. $\mathbf{W}$ is the weight matrix with dimensions $m \times d$, where m is the number of output dimensions or classes. It maps the input data $\mathbf{X}$ to the target output $\mathbf{Y}$. $\mathbf{Y}$ represents the output data matrix with dimensions $m \times n$. Each column in $\mathbf{Y}$ corresponds to the expected output for the corresponding column in $\mathbf{X}$.

$\|\mathbf{Y} - \mathbf{WX}\|_F^2$ represents the Frobenius norm of the difference between the predicted outputs $\mathbf{WX}$ and the actual outputs $\mathbf{Y}$. The Frobenius norm $\|\cdot\|_F$ is a measure of the difference between two matrices, computed as the square root of the sum of the absolute squares of its elements. In this context, $\|\mathbf{Y} - \mathbf{WX}\|_F^2$ represents the sum of squared errors between the predicted and actual outputs across all samples and all output dimensions. $\min_{\mathbf{W}}$: This notation indicates that minimize the expression with respect to $\mathbf{W}$. The goal is to find the weight matrix $\mathbf{W}$ that minimizes the difference between the predicted outputs $\mathbf{WX}$ and the actual outputs $\mathbf{Y}$, thereby achieving the best possible mapping from $\mathbf{X}$ to $\mathbf{Y}$. Conventional linear regression often employs a binary label matrix, which can be overly restrictive and may lead to overfitting by ignoring the structural information of the samples [29]. Innovative approaches have been developed to address traditional methods constraints that introduce a relaxed labeling technique [30]. These methods employ strategies that increase the separability of classes by allowing for a certain degree of label overlap, which can enhance generalization. However, these strategies often need to include consideration of the fundamental geometric arrangement within the data, potentially weakening the stability and robustness of the resulting transformation matrix. Efforts to integrate spatial relationships within data into regression models have led to graphical structures that capture the within-class and between-class connections [31]. These structures aid in condensing data representation, though they may need to provide more adaptability in the transformation matrix. Following these endeavors, an enhanced model emerged that weaves together inter-class sparsity constraints, pursuing a more discerning and pliable transformation matrix [29]. Moreover, an innovative methodology has been introduced that leverages latent space to bypass the rigidity of traditional regression techniques. This strategy involves mapping data to a deduced latent space rather than to the original label space, as delineated by Zhang et al. [32]. Complementarily, a graph reflecting similarities, informed by label data, is constructed to refine the data representation. The study proposes a dual approach that harnesses the ϵ–dragging technique in conjunction with examining within-class dispersion for soft labels.

TABLE I
Notations Used in Methodology

Notations	Descriptions
$\mathbf{S}_f \in \mathbb{R}^{d \times n_s}$	Labeled source feature matrix
$\mathbf{T}_f \in \mathbb{R}^{d \times n_t}$	Unlabeled target feature matrix
$\mathbf{S}_l \in \mathbb{R}^{c \times n_s}$	Binary source label matrix
$\mathbf{P} \in \mathbb{R}^{c \times d}$	Projection matrix
$\mathbf{L}_m \in \mathbb{R}^{c \times n_s}$	Modified label matrix
$\mathbf{V} \in \mathbb{R}^{c \times c}$	Mapping matrix
$\mathbf{B} \in \mathbb{R}^{c \times n_s}$	Constant matrix
$\mathbf{M} \in \mathbb{R}^{c \times n_s}$	Non-negative matrix
$\mathbf{L}_s \in \mathbb{R}^{n \times n}$	Similarity Laplacian matrix
$\mathbf{L}_d \in \mathbb{R}^{n \times n}$	Dissimilarity Laplacian matrix

This strategy seeks to construct a transformation matrix that exhibits superior discriminative capabilities [33]. Despite its advancements, the approach needs to include the classification structure that is inherently present within the labels. A new adaptive graph embedding method that captures the local structural nuances among data points has been developed, building upon these concepts. An innovative learning strategy complements this approach, reinforcing the discriminative power of the transformation matrix [30]. We introduce the MDERE framework to address the domain discrepancy in cross-domain SER tasks. The updated regression model, which incorporates the MDERE framework, can be formulated as follows:

$$\min_{W,R} \|WX_S - (Y_T + B \circ R)\|_F^2 + \lambda_1 \|W\|_1 + \lambda_2 \|W\|_2^2 \quad \text{s.t.} \quad R \geq 0 \tag{2}$$

In (2), the subscript 1in $\|W\|_1$ refers to the L1 norm. It is the sum of the absolute values of the elements of W. Mathematically, $\|W\|_1 = \sum |w_i|$, where w_i are the elements of W. This norm encourages sparsity in the matrix W. The subscript 2in $\|W\|_2$ refers to the L2 norm. It is the square root of the sum of the squared values of the elements of W. Mathematically, $\|W\|_2 = \sqrt{\sum w_i^2}$. This norm encourages small values in the matrix W and tends to prevent overfitting by penalizing large values.

In this model:

- X_S denotes the feature matrix from the source domain.
- Y_T represents the target domain's label matrix.
- R is the non-negative relaxation matrix, enhancing the flexibility of the label space.
- B is a binary matrix, indicating the presence or absence of features.
- λ_1 and λ_2 are regularization parameters for the elastic net, which combines $L1$ and $L2$ penalties, respectively.
- W is the transformation matrix that is optimized.

By incorporating these elements, the MDERE framework effectively aligns domain variations. It increases the generalizability of SER systems across different domains. Table I represents all the notations used in the methodology.

B. Domain Adaptation

Conventional pattern recognition algorithms typically operate under the assumption that training and test samples share an identical distribution of features. In practical scenarios, however, this assumption often needs to be revised. For instance, techniques have been developed to identify a mutual low-dimensional feature subspace, reducing the MMD between source and target domains [34]. Kan et al. proposed methods to represent source domain samples within a common subspace with those of the target domain [35]. Researchers have crafted novel deep DA frameworks that utilize attention mechanisms to assign transfer weights across various time steps [36]. Approaches such as transfer subspace learning employ relaxed transformation matrices and leverage manifold structures to aid classification tasks [25]. Recent advancements in DA include formulating pairwise regression methods and the construction of local-global graphs for inter-domain distance measurement [37]. Wang et al. also utilized iterative refinement of pseudo labels to better align samples across domains [38]. Guo et al. introduced PCA-like regularization techniques to preserve essential data features and facilitate the clustering of similarly labeled samples [39]. Despite these advancements, there remains a need for a technique that not only preserves essential data features but also enhances emotion discriminability, a gap that MDERE aims to fill. While these methods have pushed the boundaries of DA in SER, the advent of MDERE marks a significant step forward, bringing forth a comprehensive solution that not only preserves feature integrity but also accentuates discriminative learning. Specifically in the context of cross-domain SER, DA approaches have been tailored to minimize feature distribution disparities by constructing cross-domain graphs, thereby enabling the learning of more discriminative subspaces [15]. Approaches using dual distance metrics combine global measures like MMD with local metrics such as graph Laplacian to diminish distributional differences [40]. Other strategies involve using weighted intra-class difference metrics that focus on emotion categories to guide DA, and selectively adapting features in the last layer of deep neural networks to prevent potential negative transfer [41]. Additionally, preference-learning-based methods have been proposed to minimize domain discrepancies through gradient reversal and reconstruction loss [42]. Notably, many current DA methods concentrate on the similarities between domains, often at the expense of considering dissimilarity information, potentially overlooking crucial aspects during the domain alignment process. The upcoming section on methodology expounds on the technical implementation of MDERE and its optimization algorithm. Having outlined the current landscape of DA methods in SER, we now turn to a detailed description of the MDERE framework, which synthesizes and advances these techniques to address the pressing needs of cross-domain emotion recognition.

III. Methodology

The proposed methodology explicitly addresses the challenges of generalizing across diverse datasets in cross-domain SER, which simulates real-world scenarios where models encounter new, unseen speakers and varying recording conditions.

A. Preliminaries

In the context of MDERE, we define the labeled source feature matrix as $\mathbf{S}_f \in \mathbb{R}^{d\times n_s}$ and the unlabeled target feature matrix as $\mathbf{T}_f \in \mathbb{R}^{d\times n_t}$, where n_s and n_t denote the number of source and target samples, respectively, and d represents the dimensionality of the feature vectors. The binary label matrix for the source features is given by $\mathbf{S}_l \in \mathbb{R}^{c\times n_s}$, where c is the number of classes. The combined feature matrix is $\mathbf{F} = [\mathbf{S}_f, \mathbf{T}_f] \in \mathbb{R}^{d\times n}$, with $n = n_s + n_t$. For a matrix $\mathbf{Q} \in \mathbb{R}^{m\times n}$, the Frobenius norm is defined as $\|\mathbf{Q}\|_F$, and the $\ell_{2,1}$-norm as $\|\mathbf{Q}\|_{2,1}$.Our goal within MDERE is to learn a projection matrix $\mathbf{P}$ and a modified label matrix $\mathbf{L}_m$ to facilitate regression.

B. Problem Formulation

Building on Section 2(A), we propose a dual-relaxation regression model within the MDERE framework to address the inherent challenges in cross-domain SER. This model aims to learn a robust projection matrix $\mathbf{P}$ and a modified label matrix $\mathbf{L}_m$, which are capable of aligning feature distributions across domains while preserving the discriminative power of emotional expressions. The inclusion of a mapping matrix $\mathbf{V}$ with $\mathbf{L}_m$ ensures the preservation of class structure within the label space and imposes an orthogonal constraint on $\mathbf{V}$ to avoid trivial solutions. The optimization problem for dual-relaxation regression in MDERE is formulated as follows:

$$\min_{\mathbf{P},\mathbf{L}_m,\mathbf{V},\mathbf{M}} \|\mathbf{V}\mathbf{L}_m - (\mathbf{S}_l + \mathbf{B}\odot\mathbf{M})\|_F^2 + \alpha\|\mathbf{P}\mathbf{S}_f - \mathbf{L}_m\|_F^2 \quad (3)$$

subject to $\mathbf{V}^\top\mathbf{V} = \mathbf{I}$, $\mathbf{M} \geq 0$, where $\odot$ represents the element-wise product. To promote feature selectivity and model robustness, the $\ell_{2,1}$-norm of $\mathbf{P}$ is included:

$$\min_{\mathbf{P},\mathbf{L}_m,\mathbf{V},\mathbf{M}} \|\mathbf{V}\mathbf{L}_m - (\mathbf{S}_l + \mathbf{B}\odot\mathbf{M})\|_F^2 + \alpha\|\mathbf{P}\mathbf{S}_f - \mathbf{L}_m\|_F^2 + \beta\|\mathbf{P}\|_{2,1} \quad (4)$$

subject to $\mathbf{V}^\top\mathbf{V} = \mathbf{I}$, $\mathbf{M} \geq 0$.

C. Domain Alignment via Similarity and Contrastive Graphs in MDERE Framework

Given the inherent divergence in source and target data feature distributions in domain adaptation tasks, the MDERE framework introduces a sophisticated metric for quantifying inter-domain distances, incorporating similarity and dissimilarity graph-based strategies. This approach addresses traditional methodologies' limitations by leveraging similarities and differences to enhance knowledge transference across domains. The similarity graph captures shared patterns or features between the source and target domains, which facilitates the alignment of their feature spaces. This alignment is essential for improving the model's ability to transfer knowledge effectively, thereby enhancing its generalization across domains. The mathematical expression for this concept is captured as:

$$\begin{aligned}\min_{\mathbf{W}} \sum_{i,j} \left(\mathbf{W}(\mathbf{x}_i - \mathbf{x}_j)\right)^2 S_{ij} \\ = \min_{\mathbf{W}} \mathrm{Tr}\left(\mathbf{W}\mathbf{X}_s\mathbf{A}_s\mathbf{X}_s^\top\mathbf{W}^\top\right) - \mathrm{Tr}\left(\mathbf{W}\mathbf{X}_s\mathbf{S}\mathbf{X}_s^\top\mathbf{W}^\top\right) \\ = \min_{\mathbf{W}} \mathrm{Tr}\left(\mathbf{W}\mathbf{X}_s\mathbf{L}_s\mathbf{X}_s^\top\mathbf{W}^\top\right) \quad (5)\end{aligned}$$

Domain similarity matrix $\mathbf{S}$ encodes pairwise similarities between data points within the source and target domains. Each element S_{ij} represents the similarity between data points $\mathbf{x}_i$ and $\mathbf{x}_j$, facilitating domain-level alignment based on similar features.

Class dissimilarity matrix $\mathbf{D}$ encodes pairwise dissimilarities between data points based on class information, highlighting the distinctions between different classes. Each element D_{ij} represents the dissimilarity between data points $\mathbf{x}_i$ and $\mathbf{x}_j$, promoting class-level separation across domains. The **Laplacian matrix** $\mathbf{L}_s$ for the domain similarity graph is defined as $\mathbf{L}_s = \mathbf{A}_s - \mathbf{S}$, where $\mathbf{A}_s$ is the degree matrix for the domain similarity matrix. Similarly, the Laplacian matrix $\mathbf{L}_d$ for the class dissimilarity graph is defined as $\mathbf{L}_d = \mathbf{A}_d - \mathbf{D}$, enforcing structure for dissimilar data points. Both Laplacian matrices capture the structure of the respective graphs, ensuring smoothness in the data representation for similar points and emphasizing distinctions between dissimilar ones.

Objective Function: The MDERE framework seeks to learn a transformation matrix $\mathbf{W}$ that minimizes the distance between similar data points across domains while maximizing the separation between dissimilar points based on class in the transformed space. This is expressed through trace Tr operations, which summarize the diagonal elements of the matrices to compute the sum of squared differences efficiently, weighted by $\mathbf{S}$ and $\mathbf{D}$, respectively. The similarity and dissimilarity graphs play a critical role in MDERE by ensuring that $\mathbf{W}$ aligns similar data points across domains and maximizes the separation between dissimilar points across classes, enhancing the robustness of domain adaptation. Labels from the source domain are used to train a classifier that predicts provisional labels for the target domain, which are recursively refined using the projection matrix $\mathbf{W}$. The final class dissimilarity matrix is constructed using the source labels and the updated pseudo-labels for the target domain.

By integrating (5) (representing domain similarity) with (6) (representing class dissimilarity), we develop a comprehensive distance metric essential for optimizing the domain adaptation process within the MDERE framework. This unified approach leads to the following combined loss function:

$$\begin{aligned}
&\min_{\mathbf{W}} \sum_{i,j} \left(\mathbf{W}(\mathbf{x}_i - \mathbf{x}_j)\right)^2 D_{ij} \\
&= \min_{\mathbf{W}} \operatorname{Tr}\left(\mathbf{W}\mathbf{X}_d\mathbf{A}_d\mathbf{X}_d^\top\mathbf{W}^\top\right) \quad - \operatorname{Tr}\left(\mathbf{W}\mathbf{X}_d\mathbf{D}\mathbf{X}_d^\top\mathbf{W}^\top\right) \\
&= \min_{\mathbf{W}} \operatorname{Tr}\left(\mathbf{W}\mathbf{X}_d\mathbf{L}_d\mathbf{X}_d^\top\mathbf{W}^\top\right) \qquad (6)
\end{aligned}$$

$$\min_{\mathbf{W}} \operatorname{Tr}\left(\mathbf{W}\mathbf{X}(\mathbf{L}_s - \lambda\mathbf{L}_d)\mathbf{X}^\top\mathbf{W}^\top\right), \qquad (7)$$

Where $\mathbf{L}_s$ and $\mathbf{L}_d$ are the laplacian matrices for similarity and dissimilarity respectively, and λ is the balance parameter.

Incorporating this with the dual-relaxation regression framework from (3) generates the terminal objective function for our MDERE methodology:

$$\begin{aligned}
\min_{\mathbf{W},\mathbf{U},\mathbf{V},\mathbf{M}} &\left\{\|\mathbf{V}\mathbf{U} - (\mathbf{Y}_s + \mathbf{B}\odot\mathbf{M})\|_F^2 + \alpha\|\mathbf{W}\mathbf{X}_s - \mathbf{U}\|_F^2\right. \\
&\left. +\beta\|\mathbf{W}\|_{2,1} + \gamma\operatorname{Tr}\left(\mathbf{W}\mathbf{X}(\mathbf{L}_s - \lambda\mathbf{L}_d)\mathbf{X}^\top\mathbf{W}^\top\right)\right\}, \qquad (8)
\end{aligned}$$

In (8), $\mathbf{U}$ represents the latent variable that serves as an intermediate mapping between the projected feature space $\mathbf{W}\mathbf{X}_s$ and the final output $\mathbf{V}\mathbf{U}$. It captures the relationship between the transformed input features and the model's predicted outputs, allowing for a more flexible and robust representation of the underlying data structure. Subject to the orthogonality condition $\mathbf{V}^\top\mathbf{V} = \mathbf{I}$, and non-negativity of $\mathbf{M}$. Here, α, β, and γ are regularization coefficients that calibrate fitting error, feature selection, and domain adaptation. To address the optimization of (8) within our study, we have devised an iterative optimization strategy that is both straightforward and potent. The optimization challenge presented by (8) stems from the inclusion of the $\ell_{2,1}$-norm, a non-smooth entity that defies direct optimization. To circumvent this complexity, we transform the $\ell_{2,1}$-norm of $\mathbf{W}$ into a more tractable expression as delineated in prior research [12] :

$$\|\mathbf{W}\|_{2,1} = 2\operatorname{Tr}(\mathbf{W}^\top\mathbf{Z}\mathbf{W}) \qquad (9)$$

Where $\mathbf{Z} = [\zeta_{ii}]$ is a diagonal construct, with ζ_{ii} computed via $\zeta_{ii} = \frac{1}{2\|\mathbf{w}_i\|_2}$, and $\mathbf{w}_i$ denotes the i-th row vector of $\mathbf{W}$.

Observing that $\|\mathbf{w}_i\|_2$ may verge on zero, which could induce non-differentiability in our function, we introduce an infinitesimal constant v to ensure stability, thus reformulating ζ_{ii} as:

$$\zeta_{ii} = \frac{1}{2\sqrt{\|\mathbf{w}_i\|_2^2 + v}}.$$

With $\widetilde{\mathbf{Y}}_s = \mathbf{Y}_s + \mathbf{B}\odot\mathbf{M}$, we recast (8) into an optimization problem represented by:

$$\begin{aligned}
\mathcal{L} = &\|\mathbf{V}\mathbf{U} - \widetilde{\mathbf{Y}}_s\|_F^2 + \alpha\|\mathbf{W}\mathbf{X}_s - \mathbf{U}\|_F^2 \\
&+ \beta\operatorname{Tr}(\mathbf{W}^\top\mathbf{Z}\mathbf{W}) + \gamma\operatorname{Tr}(\mathbf{W}\mathbf{X}(\mathbf{L}_s - \lambda \\
&\mathbf{L}_d)\mathbf{X}^\top\mathbf{W}^\top) + \eta\operatorname{Tr}(\mathbf{V}^\top\mathbf{V} - \mathbf{I}). \qquad (10)
\end{aligned}$$

The optimization procedure is an iterative one where each variable is updated in turn while the rest are held static. The iterations are as follows: 1. **Update $\mathbf{W}$**: With other variables held constant, the function for $\mathbf{W}$ simplifies to:

$$\begin{aligned}
\mathcal{L}(\mathbf{W}) = &\alpha\|\mathbf{W}\mathbf{X}_s - \mathbf{U}\|_F^2 + \beta\operatorname{Tr}(\mathbf{W}^\top\mathbf{Z}\mathbf{W}) \\
&+ \gamma\operatorname{Tr}(\mathbf{W}\mathbf{X}(\mathbf{L}_s - \lambda\mathbf{L}_d)\mathbf{X}^\top\mathbf{W}^\top). \qquad (11)
\end{aligned}$$

- The derivative of $\mathcal{L}$ with respect to $\mathbf{W}$ yields the solution:

$$\mathbf{W} = \left(\alpha\mathbf{U}\mathbf{X}_s^\top\right)\left(\alpha\mathbf{X}_s\mathbf{X}_s^\top + \beta\mathbf{Z} + \gamma\mathbf{X}(\mathbf{L}_s - \lambda\mathbf{L}_d)\mathbf{X}^\top\right)^{-1}. \qquad (12)$$

- Update $\mathbf{U}$: Fixing the other parameters, the problem concerning $\mathbf{U}$ becomes:

$$\mathcal{L}(\mathbf{U}) = \|\mathbf{V}\mathbf{U} - \widetilde{\mathbf{Y}}_s\|_F^2 + \alpha\|\mathbf{W}\mathbf{X}_s - \mathbf{U}\|_F^2. \qquad (13)$$

  Setting the derivative of $\mathcal{L}$ with respect to $\mathbf{U}$ to zero, we deduce the solution for $\mathbf{U}$:

$$\mathbf{U} = \left(\mathbf{V}^\top\mathbf{V} + \alpha\mathbf{I}\right)^{-1}\left(\alpha\mathbf{W}\mathbf{X}_s + \mathbf{V}^\top\widetilde{\mathbf{Y}}_s\right). \qquad (14)$$

- Update $\mathbf{V}$: Maintaining other variables stationary, the update for $\mathbf{V}$ results from the derivative of $\mathcal{L}$ with respect to $\mathbf{V}$:

$$\mathbf{V} = \left(\widetilde{\mathbf{Y}}_s\mathbf{U}^\top\right)\left(\mathbf{U}\mathbf{U}^\top + \eta\mathbf{I}\right)^{-1}. \qquad (15)$$

Algorithm 1: MDERE Optimization Process.

Require: Source feature matrix $\mathbf{S}_f$, target feature matrix $\mathbf{T}_f$, source label matrix $\mathbf{S}_l$.
Ensure: Projection matrix $\mathbf{P}$.
1: Initialize $\mathbf{P}$ using PCA for dimensionality reduction.
2: Initialize $\mathbf{L}_m = \mathbf{S}_l$, $\mathbf{M}$, and $\mathbf{V}$ as identity matrices.
3: **while** not converged **do**
4: Update $\mathbf{P}$ by minimizing the objective function.
5: Update $\mathbf{L}_m$ using the current $\mathbf{P}$.
6: Update $\mathbf{V}$ to maintain orthogonality (refer to (15)).
7: Update $\mathbf{M}$ through non-negative matrix factorization (refer to (16)).
8: Check for convergence criteria.
9: **end while**
10: **return** $\mathbf{P}$.

- Update $\mathbf{M}$: With the remaining parameters immobilized, we define $\mathbf{E} = \mathbf{Y}_s - \mathbf{VU}$ and determine $\mathbf{M}$ via:

$$\mathbf{M} = \max(\mathbf{B} \odot \mathbf{E}, 0). \tag{16}$$

This iterative cycle is repeated until convergence is attained. These sequential optimization steps are pivotal to the deployment of the proposed MDERE methodology. This optimization problem is tackled iteratively, updating one variable at a time while holding the others constant. The detailed steps for updating the projection matrix $\mathbf{P}$, the modified label matrix $\mathbf{L}_m$, the mapping matrix $\mathbf{V}$, and the non-negative matrix $\mathbf{M}$ are provided in Algorithm 1, which outlines the MDERE methodology. Algorithm 1: MDERE Optimization Process

The algorithm is considered to have converged when the change in the objective function between iterations falls below a predetermined threshold, ensuring stability in the solution. The computational complexity of the MDERE algorithm primarily arises from its iterative optimization steps, matrix multiplications, and the construction of similarity and dissimilarity Laplacian matrices. Given the iterative updates of matrices such as $\mathbf{W}$, $\mathbf{U}$, $\mathbf{V}$, and $\mathbf{M}$, along with operations like Frobenius and $\ell_{2,1}$-norm calculations, the complexity is likely polynomial concerning the input data dimensions. This polynomial complexity, particularly with the iterative refinement of the projection matrix $\mathbf{W}$ and the employment of graph-based domain alignment, indicates that MDERE is computationally feasible for large-scale datasets. Through iterative optimization, MDERE achieves convergence efficiently, making it suitable for real-world applications where computational resources may be limited.

IV. Experiments

Our experiments are designed to evaluate the effectiveness of the proposed MDERE framework in addressing the challenges of cross-domain SER. By utilizing diverse datasets with different languages, recording conditions, and emotional expressions, we aim to demonstrate our approach's robustness and generalization capabilities.

A. Experimental Setup

This segment delineates the datasets utilized, the scope of cross-domain tasks, benchmarking methods, and the configuration of parameters employed in our empirical evaluation.

1) Datasets: Our empirical analysis harnesses the following corpora as detailed in Table II:

- *EMO-DB (ED) [43]:* This corpus encompasses 494 speech excerpts from 10 German speakers, encapsulating a spectrum of human emotive states such as anger, boredom, disgust, fear, happiness, sadness, and a neutral state. The recordings are acted and are in German. The total duration of the dataset is approximately 25 minutes.
- *CVE (CV) [44]:* Comprising 874 meticulously annotated speech samples spanning seven emotional categories: anger, boredom, disgust, fear, happiness, sadness, and neutral, this dataset is voiced by four native Mandarin speakers. The recordings are acted, and the dataset is in Mandarin. The total duration is approximately 46 minutes.
- *IEMOCAP (IP) [45]:* Documented by a cohort of ten actors, this compilation features five emotional expressions: frustration, sadness, happiness, anger, and neutrality. The recordings are both acted and improvised, and the dataset is in English. The total duration is approximately 12 hours.
- *TESS (TS) [46]:* A collection of utterances recorded by two actresses, TESS includes readings of words infused with seven emotional nuances: anger, disgust, fear, happiness, surprise, sadness, and a neutral tone. The recordings are acted and are in English. The total duration of the dataset is approximately 1 h and 10 minutes.

B. Segmentation and Cross-Corpus Methodology Clarification

Our experiments segmented datasets based on speaker identity to rigorously evaluate the model's ability to generalize across varying speaker characteristics. Each dataset was divided into ten segments, with 70% used for training, 15% for validation, and 15% for testing, ensuring that each set included data from 26 distinct speakers. The validation set was used for hyperparameter tuning and to prevent overfitting through early stopping. This approach simulates real-world scenarios where the model encounters new, unseen speakers. Although 70% of the target dataset was used for fine-tuning, the primary training was conducted on a separate source corpus, preserving the cross-corpus nature of the experiments. Pre-processing steps included noise reduction, silence trimming, and normalization to zero mean and unit variance, with all audio samples standardized to a fixed length of 3 seconds and resampled to 16kHz.

C. Experimental Procedures

Cross-domain SER challenges are tackled using a randomized selection from four distinct datasets. By designating each as the source and target domain alternately, we arrive at 12 unique cross-domain experiments such as ED $\rightarrow$ CV, ED $\rightarrow$ IP, ED $\rightarrow$ TS, etc. These experiments involve training on the source dataset coupled with part of the target dataset, with the remainder of the target dataset reserved for evaluation. The datasets were chosen

TABLE II
DETAILED INFORMATION OF DATASETS USED IN CROSS-CORPUS

Dataset	Language	Recording Type	Total Speakers	Training Set (70%)	Test Set (30%)	Total Utterances	Total Duration (HH:MM:SS)
EMO-DB (ED)	German	Acted	10	7 speakers	3 speakers	535	00:25:00
CVE (CV)	Mandarin Chinese	Induced	4	3 speakers	1 speaker	200	00:15:00
IEMOCAP (IP)	English	Induced	10	7 speakers	3 speakers	5,531	12:03:00
TESS (TS)	English	Acted	2	1 speaker	1 speaker	2,800	01:16:00

based on diversity of emotional content, ensuring a balanced representation of emotions to prevent bias in the training and testing phases. Our experiments employ four universal emotions: anger, neutral, happy, and sad present across all datasets.

D. Comparative Analysis

Our proposed approach is benchmarked against several recent state-of-the-art methods specifically designed for speech emotion recognition with domain adaptation, such as Domain Adversarial Neural Networks (DANN) [20], Adversarial Discrepancy Learning (ADL) [47], and Few-shot Learning with Domain Adaptation [21]. All these methods, including our MDERE framework, require careful hyperparameter tuning to achieve optimal results. For instance, DANN uses adversarial training to align feature distributions between domains, necessitating fine-tuning of key parameters. Similarly, ADL requires careful tuning to ensure stability during training in complex tasks. Few-shot Learning with Domain Adaptation also depends on fine-tuning to optimize performance with limited target data. While these methods offer advancements in domain adaptation for speech emotion recognition, they face notable limitations.

- **DANN** often struggles with stability and convergence when the domain shift is significant, particularly in emotion recognition tasks, due to the high variability in emotional expressions. Additionally, the adversarial training process in DANN requires extensive tuning of hyperparameters, which can be computationally expensive and time-consuming.
- **ADL**, which focuses on semantic consistency, faces difficulties maintaining stability during training, particularly in complex emotional expressions. This results in less reliable domain adaptation, especially when handling induced emotions.
- **Few-shot Learning** is designed for limited target data scenarios but can be severely constrained by the quality and variability of the available samples, limiting its generalization to broader applications.

Our MDERE framework addresses these limitations by leveraging graph-based similarity and dissimilarity learning strategies, which:

- Reduce the reliance on hyperparameter tuning by aligning domain discrepancies more effectively through graph-based methods.
- Ensure better stability and convergence across diverse tasks and emotional expressions, even in challenging cross-domain scenarios.
- Improve generalization to unseen target domains by incorporating similarity and dissimilarity learning, which better captures the nuances of emotional expressions across different datasets.

TABLE III
THE LOW-LEVEL DESCRIPTORS (LLDS) EMPLOYED IN OUR EXPERIMENTS

Descriptors	Number of Features
MFCC	648
LSP frequency	300
Log Mel frequency band	340
Voicing probability	45
Loudness	45
F0 envelope	45
F0	42
Shimmer	42
Jitter	42
Jitter consecutive frame pairs	42
F0 number of onsets	2
Turn duration	2

Despite the careful tuning needed for all methods, MDERE consistently demonstrates superior generalization capabilities across multiple datasets, outperforming other approaches in various cross-domain emotion recognition tasks, as shown in Tables IV and V.

MDERE consistently outperforms the benchmark in low-level and deep feature tasks, indicating its strength as a domain adaptation technique. In Fig. 2, we present confusion matrices for each of the 12 cross-domain tasks to provide a comprehensive analysis of the MDERE model's performance. Each task represents a unique challenge rooted in the distinct characteristics of the source and target domains. Below, we describe the key differences and challenges that justify the inclusion of all tasks:

- *Linguistic Diversity:* The datasets cover different languages (e.g., German, Mandarin, English), presenting challenges in emotion recognition due to varying phonetic patterns. MDERE effectively handles these variations.
- *Recording Conditions:* The datasets differ in recording environments, from controlled studio settings to more spontaneous conditions. MDERE adapts well to these differences.
- *Emotional Expression Types:* Some datasets feature acted emotions, while others include induced emotions, which are subtler. MDERE addresses these expressiveness variations.
- *Speaker Characteristics:* The number of speakers varies across datasets, requiring models to generalize well. MDERE demonstrates strong adaptability to this variability.

E. Advanced Domain Adaptation Methods

In addition, we compare MDERE to an array of advanced DA methods:

- *Joint Distribution Adaptation (JDA) [14]:* Aims to align both the marginal and conditional distributions of the source and target domains by finding a shared feature space.

TABLE IV
RECOGNITION ACCURACY (%) OF MDERE AND COMPARISON ALGORITHMS USING LOW-LEVEL FEATURES

Tasks	PCA	JDA	TCA	TSDSL	JPDA	DAPCA	TDLR	MDERE
ED → CV	47.51	52.35	52.95	54.78	59.76	63.17	69.87	68.58
ED → IP	46.05	46.05	49.21	60.37	48.19	42.57	51.17	50.26
ED → TS	50.20	47.46	47.66	59.51	54.09	58.96	62.83	64.22
CV → ED	58.50	53.17	65.74	58.03	65.01	67.70	77.31	79.29
CV → IP	45.60	49.85	46.26	49.54	48.87	56.36	51.93	54.16
CV → TS	60.31	62.65	64.42	55.46	61.44	66.34	59.95	61.53
IP → ED	31.34	43.51	61.27	48.38	58.86	47.44	57.07	63.93
IP → CV	35.49	46.36	35.49	50.59	53.30	46.99	48.20	60.73
IP → TS	33.08	58.78	48.49	51.95	52.13	56.85	53.93	64.57
TS → ED	55.11	59.98	64.81	51.32	60.77	65.74	67.68	64.81
TS → CV	59.56	60.17	60.17	55.37	62.40	60.17	63.80	68.00
TS → IP	48.12	50.07	47.95	49.77	49.00	47.63	49.22	58.75
Average	47.62	52.54	53.79	53.78	56.15	56.72	59.44	63.23

TABLE V
DEEP FEATURES, EXTRACTED USING MDERE, OFFER A HIGH-DIMENSIONAL REPRESENTATION THAT ENCAPSULATES COMPLEX PATTERNS WITHIN SPEECH DATA, INSTRUMENTAL FOR SER

Tasks	PCA	JDA	TCA	TSDSL	JPDA	DAPCA	TDLR	DAAN*	MRAN*	BNM*	MDERE
ED → CV	51.17	58.25	53.52	56.49	54.86	49.97	61.82	38.74	61.14	51.66	63.05
ED → IP	41.58	47.09	41.99	45.67	48.63	38.37	48.96	47.09	47.00	48.30	51.29
ED → TS	62.65	62.06	54.26	56.03	62.61	63.48	70.79	54.12	66.28	49.73	72.96
CV → ED	63.48	61.70	60.74	57.89	62.38	67.78	63.56	70.98	64.02	64.35	68.32
CV → IP	37.71	47.86	38.44	44.83	46.92	49.54	45.78	40.49	46.79	39.99	46.58
CV → TS	49.58	60.26	51.54	52.14	59.55	57.54	54.23	56.79	69.84	53.37	60.87
IP → ED	44.20	35.11	41.78	66.48	63.92	63.56	65.34	66.24	64.02	70.08	63.56
IP → CV	33.39	46.99	55.92	49.92	56.02	60.21	62.45	44.67	54.97	54.00	65.99
IP → TS	36.75	40.26	51.54	52.44	52.77	61.44	62.03	44.63	48.88	51.03	67.78
TS → ED	46.50	61.70	51.29	53.15	54.72	56.11	58.83	69.72	59.92	64.02	63.56
TS → CV	55.87	64.78	60.06	53.24	57.49	58.25	60.66	44.89	46.66	51.66	64.22
TS → IP	35.88	46.58	41.51	47.32	45.56	45.10	45.48	42.43	45.44	42.07	48.01
Average	46.53	52.74	50.19	52.93	55.42	55.89	58.30	51.76	56.25	53.33	61.33

TABLE VI
RECOGNITION ACCURACY (%) OF MDERE WITH DIFFERENT FEATURE SETS

Feature Set	**ED → CV**	**ED → IP**	**ED → TS**	**CV → ED**	**CV → IP**	**CV → TS**	**Average**
OpenSMILE (LLD)	68.58	50.26	64.22	79.29	54.16	61.53	63.23
ResNet-50 (Deep)	63.05	51.29	72.96	68.32	46.58	60.87	61.33
Hubert	72.34	54.12	68.45	81.29	59.11	65.89	66.87
Wav2vec	74.58	55.76	70.13	82.47	60.33	67.21	68.41

- *Transfer Sparse Discriminant Subspace Learning (TSDSL) [12]:* Focuses on finding a subspace where the source and target domain data are both sparse and well separated by class.
- *Joint Probability Distribution Adaptation (JPDA) [48]:* Aligns the joint probability distribution of feature and label spaces between domains for better generalization.
- *Domain Adaptation Principal Component Analysis (DAPCA) [49]:* An extension of PCA tailored for domain adaptation, focusing on finding a discriminative subspace for the target domain.
- *Transferable Discriminant Linear Regression (TDLR) [49]:* Applies linear regression for domain adaptation by learning a transferable transformation.
- *DAAN [50]:* A deep learning-based approach that uses attention mechanisms to focus on more transferable feature representations.
- *Multi-Representation Adaptation Network (MRAN) [51]:* Utilizes multiple feature representations to capture different aspects of the data and adapts them across domains.
- *Batch Nuclear-Norm Maximization (BNM) citeDBLP: journals/corr/abs-2003-12237:* Maximizes the batch nuclear norm to encourage learning more discriminative batch-wise features, useful in dealing with batch effects.

To ensure a fair and consistent comparison, we trained and evaluated all methods under identical experimental conditions. We implemented each algorithm following the standard procedures and hyperparameters described in their original publications. We used kernelized versions where appropriate for methods such as JDA, TCA, and BDA. Deep learning-based methods like DAAN and MRAN were implemented using TensorFlow and PyTorch, replicating the original network architectures and loss functions. Hyperparameters were optimized through 5-fold cross-validation within the training set. The

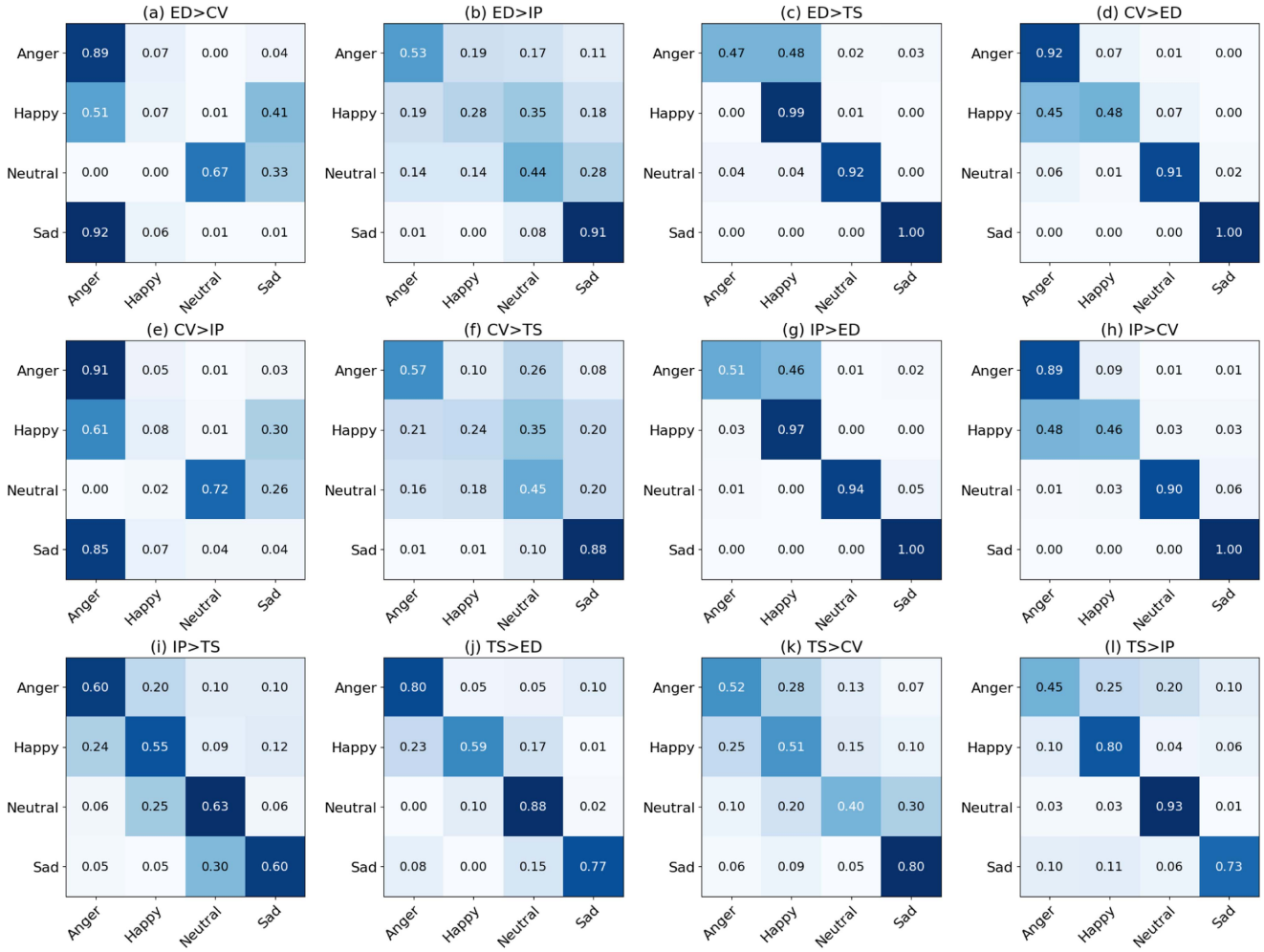

Fig. 2. The MDERE confusion matrix with low-level features displays actual labels horizontally and predicted labels vertically.

Adam optimizer with a learning rate of 0.001 was consistently used across all methods, with early stopping applied based on validation loss, allowing patience of 10 epochs to prevent overfitting. Performance metrics, including accuracy, precision, recall, and F1-score, were computed to ensure a robust and unbiased assessment of each method's performance. This uniform evaluation framework ensured that all methods were assessed equitably, allowing for a fair comparison of their effectiveness. Across all evaluations, the MDERE method demonstrates exceptional capability in cross-domain SER tasks, proving itself to be a robust method for handling diverse scenarios and complex feature representations.

F. Parameter Configuration

In our experiments, the openSMILE toolkit [52] is utilized to extract a feature set of 1582 dimensions from the INTERSPEECH 2010 paralinguistic challenge dataset. This feature set includes 34 core LLDs, such as pitch (F0), Mel-frequency cepstral coefficients (MFCCs), line spectrum pairs (LSPs), and loudness metrics. Additionally, each LLD is accompanied by its delta coefficient, adding up to 34 dynamic features. By applying 21 statistical functions to these LLDs, a 14-dimensional feature subset is derived. Furthermore, 19 statistical functions are employed on four high-frequency LLDs and their delta coefficients to yield a 152-dimensional feature subset. The dataset is augmented with two additional dimensions for the speaker's pitch and speech onset duration. Detailed attributes of these LLDs are delineated in Table III. To ascertain the efficacy of our MDERE model, we also experiment with 2048-dimensional deep features, which are extracted via Mel spectrograms using the ResNet-50 model. The deep feature experiments are conducted under the same protocol as the low-level feature set, and the performance of MDERE is benchmarked against three state-of-the-art deep DA methods. In addition to the openSMILE toolkit, we also incorporated recent advanced speech representations, specifically Hubert [53] and Wav2vec [54] features, known for their superior performance in various speech processing tasks. Hubert features were extracted using the Hubert model [53], which learns speech representations through a masked language modeling objective on the speech signal. Wav2vec features were obtained using the Wav2vec 2.0 model [54], which also leverages self-supervised learning to capture rich speech representations. These features were extracted using pre-trained models from the Fairseq library. Incorporating Hubert and Wav2vec features aimed to enhance the robustness and accuracy

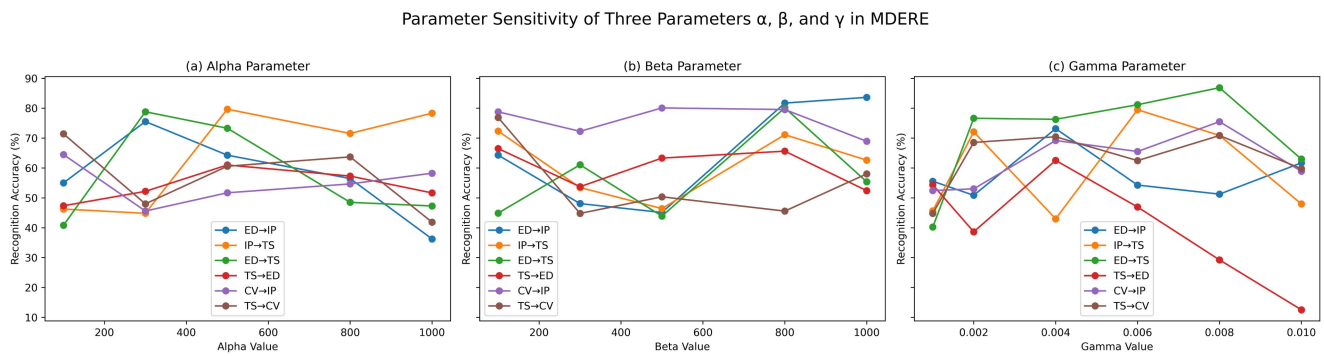

Fig. 3. Parameter sensitivity analysis for α, β, and γ.

of our MDERE model in speech emotion recognition tasks by leveraging their ability to capture nuanced speech patterns and contextual information. Due to the varying feature distributions in the source and target datasets, cross-validation is not a viable strategy for optimizing the MDERE model parameters. Therefore, we resort to a grid search in the parameter space, with candidates spanning $\{0.001, 0.01, 0.1, 1, 10, 100, 1000\}$. The number of nearest neighbors, p, is empirically set to 5. For PCA-related algorithms, we retain 98% of the energy in the principal components. We identified the optimal parameters for our MDERE model through the grid search process. In our experiments, these parameters significantly improved recognition accuracy, achieving an average of 63.23% with low-level features and 61.33% with deep features, surpassing other benchmarked algorithms. These results quantitatively demonstrate the efficacy of MDERE in domain adaptation for speech emotion recognition tasks.

Furthermore, when using Hubert and Wav2vec features, the model showed an additional improvement in recognition accuracy, highlighting the effectiveness of these modern representations. The Hubert-based MDERE model achieved an average accuracy of 66.87%, while the Wav2vec-based model reached 68.41%, outperforming the baseline and demonstrating the robustness of MDERE across different feature extraction methodologies. These experiments underscore the versatility of the MDERE framework in adapting to various high-quality speech representations.

For Hubert features, the speech signal x is processed through the Hubert model to obtain a feature representation h:

$$h = \text{Hubert}(x) \tag{17}$$

For Wav2vec features, the speech signal x is processed through the Wav2vec 2.0 model to obtain a feature representation w:

$$w = \text{Wav2vec2}(x) \tag{18}$$

Both Hubert and Wav2vec models are trained on large-scale unlabeled speech data using a masked language modeling objective, allowing them to capture rich contextual information from the speech signal.

G. Analysis of Recognition Results

Tables IV and V present the recognition accuracies of our MDERE model alongside competing algorithms, with the best outcomes being highlighted. Insights derived from these results are as follows: Our MDERE model consistently outperforms all others in both tables, improving upon the second-best method, TDLR, by 3.79% and 3.03%, respectively. This demonstrates the effectiveness of our approach. DA techniques surpass PCA as the latter only addresses single-domain issues and lacks generalization to other domains. DA methods consider the source-target domain discrepancy, thereby excelling in cross-domain tasks. Regression-based methods, including TDLR and our MDERE, achieve better performance than other DA techniques, likely because they leverage label information and refine pseudo-labels, facilitating the learning of the transformation matrix. Compared with deep DA methods, our MDERE model shows superior performance, indicating its robustness in handling deep features. Additionally, confusion matrices for each task group (see Fig. 2) indicate that emotions such as anger and sadness are recognized more accurately, possibly due to their distinctive nature. In contrast, happy and neutral are more subtle and often misclassified.

H. Parameter Sensitivity Analysis

The optimal values for parameters α, β, and γ underscore the importance of feature selection, balance between domain adaptation and fitting error. An investigation into the sensitivity of the MDERE model to its parameters α, β, and γ in (8) reveals: The model achieves the highest recognition accuracy at $\alpha = 200$ (Fig. 3(a)). The optimal performance of MDERE is observed at $\beta = 1000$ (Fig. 3(b)). A smaller value of γ within the range of 0.002 to 0.004 enhances cross-domain classification tasks (Fig. 3(c)).

I. Ablation Analysis

The ablation study for MDERE, illustrated in Fig. 4, shows: A significant decrease in recognition accuracy when α is set to 0, emphasizing the importance of a discriminative transformation matrix. Varied reductions in accuracy upon setting β to 0, demonstrating the efficacy of the $\ell_{2,1}$-norm constraint in

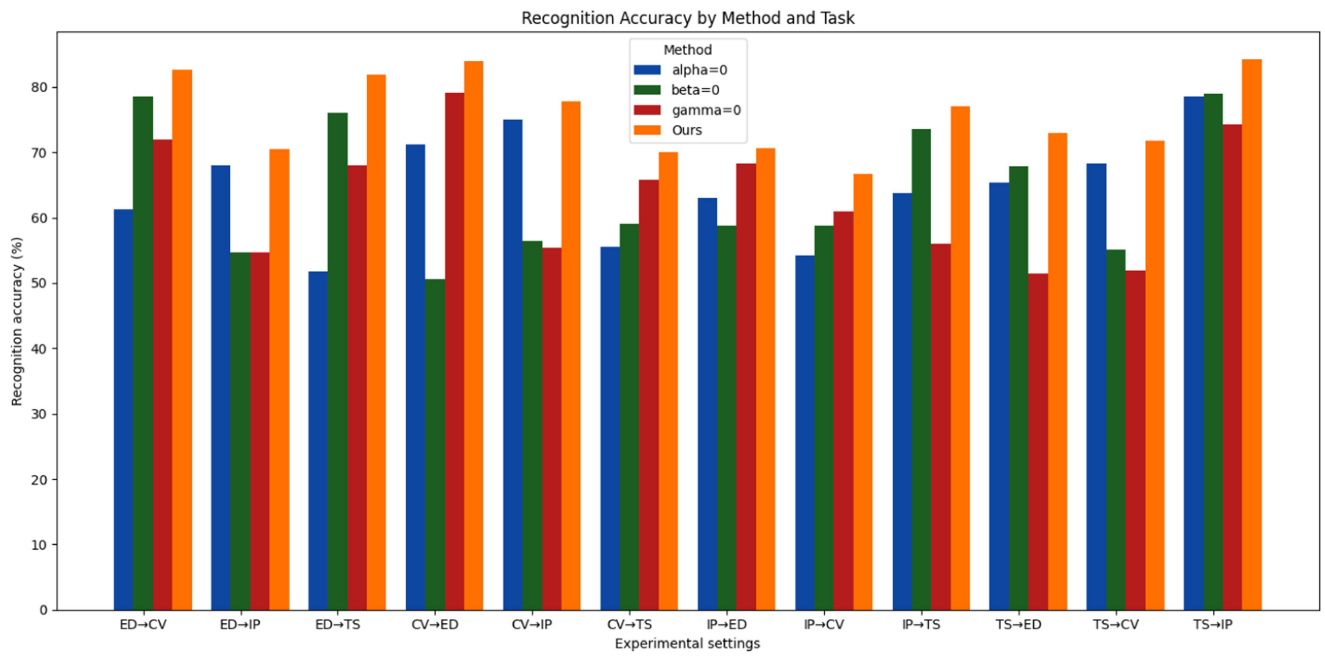

Fig. 4. Ablation study results for MDERE.

improving model robustness and reliability. A slight decline in accuracy when γ is set to 0, indicating the necessity of the graph-based distance metric.

J. Computational Complexity Analysis

The computational complexity of the MDERE algorithm primarily depends on the matrix operations within each iteration. The most computationally intensive step is the update of the projection matrix $\mathbf{P}$, which involves matrix multiplication and inversion operations. Assuming the feature dimension is d, source sample size is n_s, and target sample size is n_t, the complexity of updating $\mathbf{P}$ is $O(d^2(n_s + n_t))$. Given that the algorithm iterates k times until convergence, the overall complexity is $O(kd^2(n_s + n_t))$. This reflects the scalability of MDERE to larger datasets and informs the practical computational resources required.

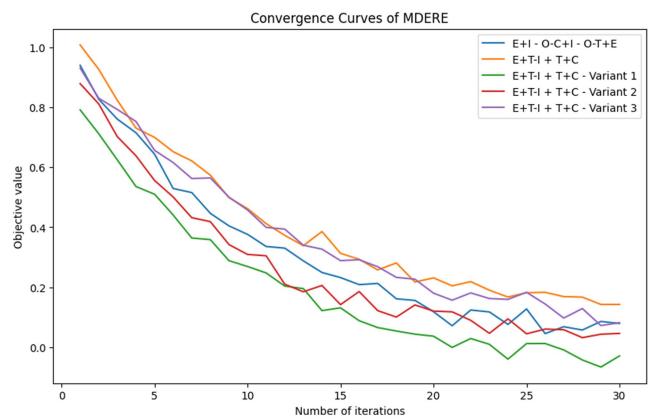

Fig. 5. Convergence analysis for MDERE.

K. Convergence Analysis

This section presents a detailed convergence analysis of the MDERE method, as depicted in Fig. 5. The convergence curves for different variants of the MDERE method across multiple iterations are shown. Examination of the figure reveals that all variants exhibit a consistent downward trend in the objective value, indicating the effective convergence of the optimization process. This downward trend is critical as it demonstrates that the model is progressively moving towards an optimal solution, which is a crucial factor in ensuring the reliability and stability of the MDERE method. Most curves show significant improvement in the objective value within the first ten iterations, with the rate of decrease slowing down as the iterations progress, eventually stabilizing. Notably, the curve for the variant labeled "E+T+I + T+C - Variant 1" (green line) achieves the lowest objective value, stabilizing around iteration 20. This suggests that this variant may offer the most efficient convergence and best performance among the tested methods. The consistent decline across all methods, with no significant oscillations or increases, further validates the robustness of the MDERE method. This stable convergence behavior across different settings underscores the method's effectiveness and ensures the model can be reliably applied to a wide range of cross-domain tasks. By stabilizing the curves after approximately 20 to 25 iterations, we can confirm that the MDERE method reaches a stable solution, which is essential for its application in real-world scenarios.

These observations substantiate the proposed MDERE method's effectiveness in achieving high accuracy and maintaining stable and reliable convergence across various experimental conditions. This stability is a critical component in validating the

robustness of our approach, making the convergence analysis a necessary and integral part of our evaluation.

L. Feature Visualization

The effectiveness of the MDERE model is demonstrated through feature visualization across multiple domain adaptation methods, as shown in Fig. 6. Fig. 6(a) represents the original feature distribution (TS → CV Original), showing significant overlap between categories with poorly clustered data points. Fig. 6(b) illustrates the MDERE model (TS → CV MDERE), which achieves superior clustering with well-separated categories and well-aligned centroids, indicating effective domain adaptation. Other methods are depicted as follows: Fig. 6(c) shows the JDA method, which displays moderate separation between classes but with more dispersed clusters; Fig. 6(d) presents the TSDSL method, where reasonable class separation is achieved, but with less compact clusters and less-aligned centroids; Fig. 6(e) shows the JPDA method, which has similar performance to TSDSL, with noticeable overlap between classes and less compact clusters; Fig. 6(f) presents the BNM method, displaying moderate separation and alignment, but with less compact clusters compared to MDERE; Fig. 6(g) demonstrates the DAPCA method, which achieves moderate alignment but with more spread-out clusters; Fig. 6(h) shows the TDLR method, where significant overlap between classes is visible, and centroids are not well-separated; Fig. 6(i) displays the DAAN method, which exhibits overlapping clusters and poorly separated centroids; and Fig. 6(j) shows the MRAN method, with some compactness but still noticeable overlap between classes. While these methods show varying degrees of improvement in clustering compared to the original data, none achieve the same level of distinct category separation and compactness as MDERE. This visualization highlights MDERE's effectiveness in aligning cross-domain feature spaces, leading to better generalization and performance across domains.

V. Conclusion

This paper introduces a novel domain adaptation (DA) strategy, MDERE, designed to address the challenges of cross-domain speech emotion recognition (SER). MDERE employs an innovative dual-relaxation regression technique to derive a transformation matrix with robust discriminative capabilities. It also integrates a mapping matrix that aligns the relaxed label matrix with the structure of the original labels. To minimize the distance between source and target data, MDERE constructs tailored similarity and dissimilarity graphs. Empirical tests across multiple datasets confirm that MDERE outperforms current leading approaches in SER.

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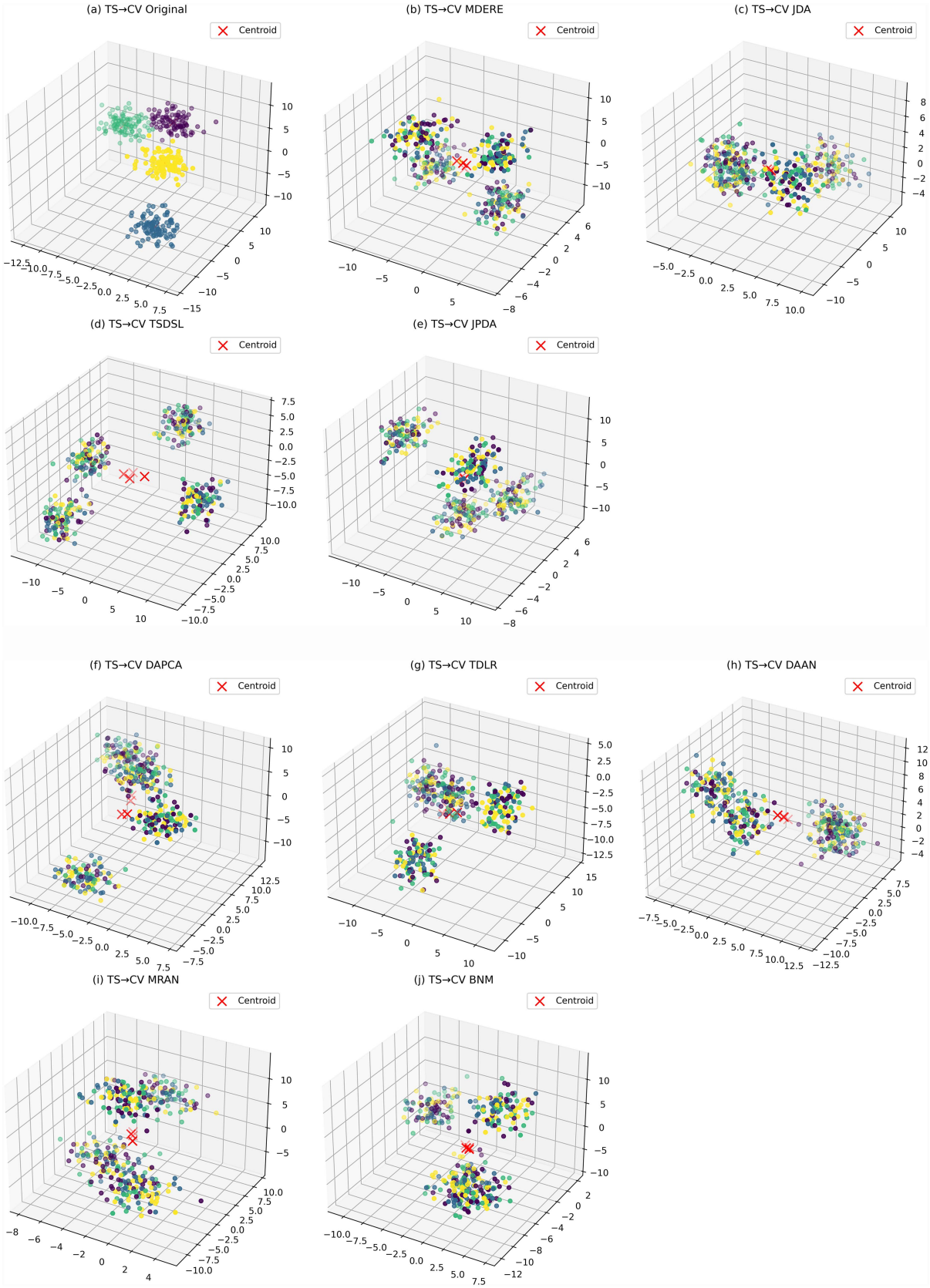

Fig. 6. Feature visualization for the TS → CV task using various domain adaptation methods. Panel (a) shows the original feature distribution, while panel (b) illustrates the MDERE model's adapted distribution with well-separated, compact clusters. Panels (c)–(j) display other methods (JDA, TSDSL, JPDA, BNM, DAPCA, TDLR, DAAN, MRAN).

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